A

PROJECT REPORT

On

An Explainable Stacking Model for Early Detection of Neurodevelopmental Disorders in Children

Submitted for partial fulfilment of

Bachelor of Science (Information Technology)

AT

AMITY INSTITUTE OF INFORMATION TECHNOLOGY

BY

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Under the Supervision of

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AMITY INSTITUTE OF INFORMATION TECHNOLOGY
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MAY 2026

AMITY UNIVERSITY
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Amity Institute of Information & Technology

CERTIFICATE

This is to certify that **Ms. Bhagyashree Sharma** bearing the Enrollment No. A21704923004 from Amity Institute of Information Technology, Amity University Rajasthan, a student of **B.Sc. (IT),** has completed the project entitled **An Explainable Stacking Model for Early Detection of Neurodevelopmental Disorders in Children** as part of the course curriculum. She completed the Project using Python tools from January-June 2026 under my supervision at Amity University Rajasthan. She has successfully completed the assigned project within the given time frame. She is a sincere and hardworking student, and her conduct during the period was commendable.

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CERTIFICATE

This is to certify that **Ms. Bhagyashree Sharma** bearing the Enrollment No. **A21704923004** from Amity Institute of Information Technology, Amity University Rajasthan, a student of **B.Sc. (IT),** has successfully completed the project entitled **'An Explainable Stacking Model for Early Detection of Neurodevelopmental Disorders in Children'** as part of the course curriculum.

Prof. Neeraj Bhargava
Amity Institute of Information Technology
Amity University Rajasthan

DECLARATION

I, Ms. Bhagyashree Sharma D/o Dinesh Kumar Sharma, a resident of Jaipur, declare that the project work incorporated in the present report entitled **"An Explainable Stacking Model for Early Detection of Neurodevelopmental Disorders in Children"** is my original work. This work (in part or whole) has not been submitted to any University for the award of a degree. I have properly checked & acknowledged the material collected from secondary sources wherever required. I am solely responsible for the originality of the entire content.

Student Name: Bhagyashree Sharma
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Extensive endeavor and blissful euphoria accompanying my task's successful completion would not be complete without expressing gratitude to the people whose guidance and encouragement helped us finish this project.

I would like to thank **Dr. Amit Kumar Singh, Assistant Professor,** Amity Institute of Information Technology, Amity University Rajasthan, who has been a perennial source of inspiration to me.

I sincerely thank **Prof. Neeraj Bhargava**, **Head of the Institute**, Amity Institute of Information Technology for his constant guidance and moral support. I also thank all faculty members of AIIT for their help during my tenure in the institute.

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ORGANIZATION PROFILE

Amity University Rajasthan

Amity University Rajasthan is a prominent private university functioning under the Amity Education Group, which ranks among the largest private educational organizations in India. Established in 2008 by the Ritnand Balved Education Foundation, the university was built with a vision to provide career-focused education while fostering a culture of research, creativity, and professional growth. Situated in Jaipur, the university enjoys the advantage of being in a rapidly developing academic and technological environment, giving students valuable exposure to both scholastic and industrial spheres.

The university follows a multidisciplinary model of education, delivering programs spanning engineering, management, law, sciences, and humanities. Its academic framework is structured to bridge theoretical understanding with hands-on application, which holds particular significance in domains like artificial intelligence, data science, and machine learning.

Campus and Infrastructure

The campus of Amity University Rajasthan is thoughtfully developed to support both academic pursuits and extracurricular growth. Spanning a vast area, the infrastructure reflects a contemporary educational setting with emphasis on practicality and efficiency. Academic blocks are outfitted with smart classrooms, digital learning platforms, and high-speed internet access, enabling smooth delivery of both conceptual and applied content.

Laboratories serve as a vital pillar of the infrastructure, particularly for technical disciplines. These facilities house up-to-date hardware and software that allow students to tackle real-world challenges. Computing labs offer access to programming platforms, machine learning environments, and data analysis tools, resources that are directly relevant to projects such as the one discussed in this report. This exposure significantly narrows the divide between academic theory and practical execution. The university library stands as another essential resource, housing an extensive collection of textbooks, research journals, scholarly papers, and digital databases. Students can access both domestic and international publications, making it an invaluable asset for literature reviews and research endeavors. Organized study spaces and efficient digital retrieval systems further streamline the process of gathering information.

Beyond academics, the campus is equipped with recreational and sports facilities, including indoor and outdoor complexes, fitness centers, and activity halls. Though often considered secondary, these amenities play a meaningful role in sustaining student productivity and psychological well-being, especially during demanding academic phases.

Residential accommodation is provided through well-maintained hostels designed to house students from various backgrounds and regions. Equipped with essential amenities and reliable security measures, the hostels offer a safe and stable living environment. This setup ensures students can access campus resources consistently, supporting continuity in their academic work.

Academic Programs

Amity University Rajasthan delivers a broad spectrum of undergraduate, postgraduate, and doctoral programs across numerous disciplines. Key faculties and departments include Engineering and Technology covering computer science, mechanical, civil, and electronics and communication engineering; Management offering MBA, BBA, and specialized tracks; Law with both integrated and standalone degree options; Applied Sciences encompassing Biotechnology, Microbiology, and Environmental Science; Arts and Humanities including English, History, Political Science, and Sociology; Commerce at both undergraduate and postgraduate levels; and Journalism and Mass Communication focusing on Media Studies and Journalism.

Every program is crafted to reflect current industry benchmarks and to prepare students with the knowledge and capabilities needed for professional advancement.

Faculty and Teaching Methodology

The teaching staff at Amity University Rajasthan comprises highly credentialed and experienced professionals, a significant number of whom hold doctoral degrees and carry substantial industry backgrounds. The pedagogical approach at AUR integrates conventional and contemporary methods, with a strong orientation toward interactive and experiential learning. This is achieved through a combination of lectures, seminars, workshops, industrial site visits, and hands-on project work. The university also leverages technology in its teaching practices, utilizing smart classrooms and online learning resources to enrich the overall educational experience.

Research and Innovation

Research occupies a central position in the academic ethos of Amity University Rajasthan. The university actively encourages both students and faculty members to engage in research activities, offering adequate resources and opportunities to support such work. Several dedicated research centers and laboratories equipped with advanced technology have been established to enable high-level inquiry. Partnerships with industries and research bodies further expand the breadth and significance of research undertaken at the institution.

Global Exposure and Collaborations

Amity University Rajasthan places considerable importance on international engagement and global collaborations. The university maintains partnerships with several recognized institutions worldwide, facilitating student and faculty exchange programs, joint research initiatives, and international internship opportunities.

Placements and Career Support

The placement division at AUR works actively toward ensuring favorable career outcomes for graduating students. It organizes a range of initiatives including campus recruitment drives, internship programs, skill-development workshops, and one-on-one career guidance sessions. The university maintains a robust network of corporate connections, with numerous reputed organizations participating regularly in its campus placement activities.

Amity University Rajasthan stands as a symbol of academic excellence and all-round student development. Through its modern facilities, wide-ranging programs, skilled faculty, and firm commitment to research and global engagement, AUR continues to shape capable professionals and future leaders.

Student Life and Extracurricular Activities

Life at Amity University Rajasthan is vibrant and dynamic, with many extracurricular activities and events. The university hosts various cultural, technical, and sports events throughout the year, allowing students to showcase their talents and develop their interests. Clubs and societies dedicated to different fields, such as robotics, drama, music, and debate, allow students to pursue their passions and engage in meaningful activities outside the classroom. allowing students.

Amity University Rajasthan stands as a beacon of quality education and holistic development. With its modern infrastructure, diverse academic programs, experienced
faculty, and strong emphasis on research and global exposure, AUR is dedicated to nurturing future leaders and professionals. The university's commitment to providing a comprehensive learning experience ensures that students are well-prepared to contribute positively to society and excel in their chosen career.

CHAPTER-1

1. INTRODUCTION

1.1 Problem Statement

Neurodevelopmental disorders represent a broad category of conditions that emerge during the developmental period and are characterized by deficits in personal, social, academic, and occupational functioning. As defined by the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, NDDs encompass Attention Deficit Hyperactivity Disorder, Autism Spectrum Disorder, Developmental Delay, Intellectual Disability, Specific Learning Disability, and Speech and Language Disorders. These conditions are not transient but are lifelong in nature and can have profound consequences on a child's cognitive development, social integration, and overall quality of life if left undetected during critical early developmental windows.

The global burden of NDDs in children is substantial. Epidemiological studies suggest that NDDs collectively affect approximately 10 to 15 percent of the pediatric population worldwide, with prevalence rates continuing to rise. Despite this growing burden, early and accurate diagnosis remains one of the most persistent challenges in pediatric healthcare. Traditional diagnostic approaches rely heavily on clinical observation, standardized assessments, and specialist availability, all of which are time-consuming, expensive, and highly subjective. The high degree of symptomatic overlap between NDDs further compounds the difficulty, often resulting in delayed or missed diagnoses precisely when early intervention would be most effective.

The inadequacy of traditional diagnostic systems has created an urgent need for scalable, automated, and data-driven approaches to NDD screening. Machine learning has emerged as a promising avenue in this regard, with studies demonstrating that ML models trained on large-scale health survey data can identify patterns associated with neurodevelopmental conditions with clinically meaningful accuracy. However, a critical limitation of most existing approaches is their exclusive focus on predictive performance, with little attention given to the transparency and interpretability of model decisions, which is a non-negotiable requirement in clinical healthcare where clinicians must be able to understand, trust, and act upon model outputs.

The challenge of class imbalance, where positive NDD cases represent a minority of the overall population, remains a significant methodological concern. Without appropriate resampling strategies, classification models tend to be biased toward the majority class, resulting in poor sensitivity for the very cases they are designed to detect. The integration of techniques such as the Synthetic Minority Oversampling Technique alongside robust ensemble methods has shown promise in addressing this challenge across various healthcare classification tasks.

In response to these gaps, this study proposes an explainable machine learning framework for the early detection of neurodevelopmental disorders in children using the 2024 National Survey of Children's Health dataset, a nationally representative survey of 51,375 children aged 0 to 17 years conducted by the United States Census Bureau. The proposed framework integrates a three-stage feature selection pipeline, SMOTE-based class balancing, a stacking ensemble classifier, and SHAP-based explainability to deliver both high predictive accuracy and clinically interpretable outputs. A risk stratification module further categorizes model predictions into low, medium, and high risk tiers to support real-world clinical decision-making.

1.2 Objectives

The primary objective of this project is to design and evaluate an explainable machine learning framework for the early detection of neurodevelopmental disorders in children. The motivation behind this objective arises from the growing burden of NDDs in the pediatric population and the inadequacy of existing diagnostic systems to meet the demand for timely, accurate, and accessible screening.

The first objective is to develop a robust data preprocessing pipeline that handles the high-dimensional NSCH dataset effectively. This involves loading and cleaning 51,375 child health records, constructing a binary NDD label from six individual disorder columns, removing data leakage sources, imputing missing values, and performing a stratified train-test split to ensure reproducibility and fairness in evaluation.

The second objective is to implement a three-stage feature selection pipeline that systematically reduces 457 raw survey features to a compact set of 40 clinically relevant variables. The three stages, namely Variance Threshold, Chi-Square testing, and Random Forest importance ranking, ensure that only features with genuine predictive value are retained for model training, improving both efficiency and interpretability.

The third objective is to address the class imbalance problem in the dataset using the Synthetic Minority Oversampling Technique. By applying SMOTE exclusively on the training set after the train-test split, the framework ensures that synthetic samples do not contaminate the test set, preserving the integrity of model evaluation.

The fourth objective is to train and compare four base machine learning classifiers, namely Logistic Regression, Random Forest, XGBoost, and Support Vector Machine, on the balanced training data and evaluate their individual performance using standard classification metrics.

The fifth objective is to implement a stacking ensemble model that combines the predictions of all four base classifiers using Logistic Regression as the meta-learner, thereby leveraging the complementary strengths of each individual model for improved and more consistent predictions.

The sixth objective is to integrate SHAP-based explainability into the framework, providing both global and local interpretability. Global SHAP analysis identifies the features most consistently driving NDD predictions across the entire test set, while local analysis explains individual predictions for specific patients, enabling clinicians to understand the reasoning behind each risk assessment.

The seventh objective is to implement a risk stratification module that categorizes each child's NDD probability into low, medium, or high risk tiers, and to develop an interactive clinical screening tool that allows practitioners to enter patient details and receive instant risk predictions with explanations.

The eighth and final objective is to conduct disorder-specific analysis by training separate models for ADHD, Autism, and Developmental Delay, comparing their individual performance and identifying disorder-specific predictive features.

The overall objective of this project is to build an explainable machine learning framework that can accurately detect neurodevelopmental disorders in children at an early stage. This is achieved by preprocessing a large-scale national health survey dataset, selecting the most relevant features, addressing class imbalance, and training an ensemble of machine learning models. The framework goes beyond just prediction by incorporating SHAP-based explainability and a risk stratification module, ensuring that its outputs are transparent and actionable for clinical use. Additionally, disorder-specific models for ADHD, Autism, and Developmental Delay are developed to provide deeper and more targeted diagnostic insights.

1.3 Research Questions

The research questions defined in this study are aimed at critically evaluating whether the proposed framework effectively addresses the challenges of early NDD detection, class imbalance, high dimensionality, and clinical interpretability. The three research questions guiding this study are as follows.

RQ1: How do traditional machine learning models, namely Logistic Regression, Random Forest, XGBoost, and SVM, compare in performance for NDD detection using large-scale parent-reported survey data?

This question is motivated by the need to establish a clear performance baseline for individual classifiers on the NDD detection task. While ensemble methods are widely used in healthcare classification, the individual contributions of each base model must first be understood. The answer to this question will determine which classifiers are most suitable for the NDD detection problem and justify the selection of base models for the stacking ensemble.

RQ2: Does a stacking ensemble approach outperform individual base classifiers in detecting neurodevelopmental disorders in children?

This question addresses the core methodological contribution of the study. Stacking ensembles combine the predictions of multiple base models through a meta-learner, theoretically leveraging the complementary strengths of each individual classifier. The answer to this question will validate whether the added complexity of a stacking architecture produces a meaningful improvement over the best individual model.

RQ3: How can SHAP-based explainability enhance the clinical interpretability and trustworthiness of machine learning predictions in pediatric NDD screening?

This question addresses a critical gap in the existing literature, where most ML-based NDD detection systems function as black boxes without providing clinicians any insight into how predictions are made. The answer to this question will demonstrate whether SHAP can translate complex ensemble model outputs into clinically actionable insights that practitioners and can understand and trust.

1.4 Brief Description of the Work

This project focuses on the design, implementation, and evaluation of an explainable machine learning framework for the early detection of neurodevelopmental disorders in children. The work is motivated by the increasing prevalence of NDDs in the pediatric population and the inadequacy of traditional diagnostic approaches to meet the growing demand for timely and accessible screening.

The implementation begins with the loading and preprocessing of the 2024 National Survey of Children's Health dataset, which contains 51,375 parent-reported child health records with 457 survey features. A binary NDD label is constructed from six individual disorder columns representing ADHD, Autism Spectrum Disorder, Developmental Delay, Intellectual Disability, Learning Disability, and Speech Disorder. Preprocessing steps include removal of data leakage columns, retention of numeric features, median imputation for missing values, and an 80 to 20 stratified train-test split.

The next stage involves a three-step feature selection pipeline. First, Variance Threshold removes features with near-zero variance, reducing the feature space from 432 to 371. Second, Chi-Square testing identifies the top 80 features most statistically associated with the NDD label. Third, Random Forest importance ranking selects the top 40 features from the remaining 80, producing a compact and clinically meaningful feature set for model training.

To address the class imbalance in the training set, where 31,680 negative samples and 9,420 positive samples create a 77 to 23 percent split, SMOTE is applied exclusively on the training data. The resulting balanced training set contains 31,680 samples per class, with a total of 63,360 training samples.

Four base classifiers, namely Logistic Regression, Random Forest, XGBoost, and Support Vector Machine, are trained on the balanced data and evaluated on the held-out test set. A stacking ensemble is then constructed with these four models as the first layer and Logistic Regression as the meta-learner, trained with five-fold cross-validation.

SHAP is applied to the XGBoost model to generate a global summary plot identifying the most influential features across the test set. A risk stratification module categorizes each child's predicted NDD probability into low, medium, or high risk tiers. An interactive clinical predictor and a disorder-specific screening questionnaire are developed as practical tools for real-world deployment.

1.5 Main Contributions

The primary contributions of this work lie in the development and evaluation of an end-to-end explainable machine learning framework for early NDD detection that addresses multiple research gaps simultaneously. A key contribution of this work is the construction of a comprehensive NDD label from the NSCH 2024 dataset that covers six distinct neurodevelopmental disorders simultaneously. Unlike most existing studies that focus on a single disorder, this approach enables population-level screening that captures the full spectrum of NDDs in children, making it clinically more useful and realistic.

Another important contribution is the three-stage feature selection pipeline that systematically reduces 457 raw survey features to 40 clinically meaningful variables. This combination of Variance Threshold, Chi-Square testing, and Random Forest importance provides a statistically grounded and practically efficient method for managing high-dimensional health survey data.

The correct application of SMOTE exclusively on the training set after the train-test split represents a methodological contribution that addresses a common error in the literature where SMOTE is applied before splitting, leading to data leakage. This ensures that model evaluation reflects true generalization performance.

The stacking ensemble architecture, which combines four diverse base classifiers through a Logistic Regression meta-learner with five-fold cross-validation, provides a principled approach to combining model predictions that goes beyond simple voting or averaging.

The integration of SHAP-based explainability provides both global and local interpretability that directly supports clinical adoption. The identification of functional difficulty, memory conditions, and healthcare access as the primary predictors of NDD risk provides actionable clinical insights grounded in data.

The risk stratification module and interactive clinical screening tool represent practical contributions that bridge the gap between machine learning research and real-world healthcare deployment. These tools enable clinicians to use the framework directly without requiring any technical knowledge.

Finally, the disorder-specific analysis that trains separate models for ADHD, Autism, and Developmental Delay provides additional clinical granularity and reveals the relative difficulty of detecting each disorder individually.

Key Contributions:

- Construction of a multi-disorder NDD label covering six distinct conditions from the NSCH 2024 dataset
- A three-stage feature selection pipeline reducing 457 features to 40 clinically relevant variables
- Correct SMOTE application to prevent data leakage during class balancing
- A stacking ensemble combining Logistic Regression, Random Forest, XGBoost, and SVM with LR meta-learner
- SHAP-based global and local explainability for clinical interpretability
- A risk stratification module categorizing predictions into low, medium, and high risk tiers

- An interactive clinical screening tool with patient-level explanations
- Disorder-specific analysis for ADHD, Autism, and Developmental Delay

1.6 Remaining Sections

The remaining sections of this report are structured to provide a comprehensive understanding of the research background, methodology, experimental evaluation, and key findings of the study.
The Related Work section presents an overview of existing research in machine learning-based NDD detection, with a focus on feature selection, class imbalance handling, ensemble methods, and explainable AI. It also identifies key research gaps that the present study addresses.

The Methodology section describes the overall approach adopted in this project. It includes details about the dataset, preprocessing steps, feature selection pipeline, SMOTE application, model training, stacking ensemble construction, evaluation metrics, SHAP explainability, and risk stratification. The experimental setup outlines the tools, frameworks, and configurations used to ensure consistency and reproducibility.

The Results section presents the performance of all five models using accuracy, precision, recall, F1-score, and ROC-AUC curves. It includes confusion matrices, a model comparison table, SHAP analysis results, disorder-specific model performance, and risk categorization outputs.
The Discussion section provides a detailed interpretation of the results with respect to the three research questions, and compares the findings with existing literature.

The Conclusion section summarizes the overall work, acknowledges limitations, and suggests directions for future research.

CHAPTER-2

2. RELATED WORK

The challenge of detecting neurodevelopmental disorders early has attracted significant research attention across the fields of clinical medicine, computational neuroscience, and machine learning. Researchers have increasingly turned to data-driven approaches to support or supplement traditional diagnostic pathways, recognizing that automated screening tools could help reduce diagnostic delays and extend the reach of clinical services to populations that currently lack access to specialist care. This section reviews the existing body of research relevant to the proposed framework, covering ML-based NDD detection, feature selection for high-dimensional health data, class imbalance handling, ensemble learning in healthcare, and explainable artificial intelligence techniques.

2.1 Machine Learning for NDD Detection

The application of machine learning to the detection and prediction of neurodevelopmental disorders has gained considerable momentum over the past decade. Early work in this area focused primarily on Autism Spectrum Disorder, motivated by the availability of neuroimaging and behavioural datasets and the clear clinical need for faster diagnostic support. Support Vector Machines were among the first classifiers applied to ASD detection tasks, with studies using structural MRI data and behavioural scores to distinguish ASD from neurotypical individuals. These early studies established that ML classifiers could achieve competitive diagnostic accuracy, although small sample sizes and single-site datasets limited the generalizability of findings.

Parikh et al. [8] conducted one of the more widely cited studies on ML for ASD diagnosis, demonstrating that optimized classifiers trained on personal characteristic and behavioural data, including age, gender, family history, and communication scores, could significantly enhance the accuracy and speed of ASD identification. Their study compared multiple classifiers and found that ensemble methods consistently outperformed individual models. This finding is directly relevant to the methodology adopted in the present study, which uses a stacking ensemble built on four diverse base classifiers. The work of Parikh et al. also highlighted the importance of feature selection, noting that models trained on carefully selected subsets of features performed better than those trained on all available variables.

Rajagopalan et al. [4] presented a comprehensive review and empirical analysis of machine learning models applied to neurodevelopmental disorder prediction using electronic health records. Their study covered multiple NDDs including ADHD, ASD, and Developmental Delay, and found that gradient boosting methods, particularly XGBoost and Random Forest, consistently delivered the most reliable and generalizable results across disorder categories. Their conclusion that tree-based ensemble methods are well-suited to the structured, tabular nature of health record data directly supports the choice of XGBoost and Random Forest as base classifiers in the present framework.

A study published in the Journal of Neurodevelopmental Disorders [11] examined the potential of automated screening tools to reduce diagnostic delays in underserved communities. The authors found that ML-based preliminary screening, even when based solely on parent-reported survey data, could correctly identify a high proportion of children later confirmed with NDDs. Critically, they noted that the greatest clinical benefit came not from perfect classification accuracy but from the ability to triage large numbers of children quickly and prioritize those most likely to need specialist evaluation. This triage function is directly aligned with the risk stratification module developed in the present study.

The MDPI study [5] applied multiple machine learning classifiers to national health survey data for NDD prediction and reported accuracy rates above 88 percent for several models. The study's findings confirmed that parent-reported survey responses contain sufficient information for effective NDD screening, without requiring expensive clinical measurements or neuroimaging. However, the study did not address class imbalance or model explainability, both of which are significant limitations for clinical deployment. The present study extends this work by incorporating SMOTE and SHAP alongside the classification pipeline.

2.2 Feature Selection in High-Dimensional Health Datasets

High-dimensional health datasets present a well-known challenge for machine learning. When the number of features is large relative to the number of samples, or when many features are irrelevant or redundant, model performance can degrade through overfitting and increased computational complexity. Feature selection is therefore an essential preprocessing step in most health data ML pipelines.

The study by Han et al. [6] used a conjunctive clause evolutionary algorithm to identify ASD biomarkers from a dataset combining behavioral assessments and neuroimaging measurements. Despite the small sample size of only 28 children, the study demonstrated that reducing a large feature space to a targeted subset of clinically meaningful variables significantly improved classification accuracy and model interpretability. The study's finding that theory of mind assessments and cingulate cortex volume measurements were the most discriminative features for ASD provides a template for the kind of clinically meaningful feature identification that the present study aims to replicate using survey data.

The Frontiers study [14] on ML for ASD complicated by intellectual disability applied feature selection as a preprocessing step before training their diagnostic classifier. They reported that dimensionality reduction not only improved model accuracy but also made the resulting feature set more clinically interpretable, since the selected features tended to align with established clinical knowledge about the disorder. This finding supports the three-stage feature selection approach used in the present study.

Variance Threshold is widely used as a first-pass filter for removing uninformative features. Features with near-zero variance across all samples do not contribute meaningful information for classification, regardless of the classifier used. Chi-Square testing provides a statistically principled method for identifying features that are significantly associated with the target variable, making it particularly appropriate for binary classification tasks with categorical or ordinal features such as those found in health surveys. Random Forest importance provides a model-based measure of predictive relevance that captures non-linear interactions between features, going beyond what simple statistical tests can detect. The combination of these three methods in a sequential pipeline ensures that the final feature set is filtered from multiple complementary perspectives.

2.3 Handling Class Imbalance in Healthcare Classification

Class imbalance is a pervasive challenge in healthcare machine learning. In population-level datasets, the prevalence of any specific clinical condition is typically much lower than the general healthy population, creating datasets where the positive class is significantly underrepresented. When standard classifiers are trained on such data without correction, they tend to achieve high overall accuracy by predicting the majority class for most samples, while performing poorly on the minority class. This represents a fundamental failure for clinical screening, where detecting true positive cases is the primary objective.

The Synthetic Minority Oversampling Technique, introduced by Chawla et al. and widely adopted in the healthcare ML literature, addresses this problem by generating synthetic samples of the minority class. Unlike simple duplication, SMOTE creates new samples by interpolating between existing minority class instances in the feature space, producing examples that are plausible but not exact copies. This technique has been shown to improve minority class recall substantially in healthcare classification tasks.

Sowjanya and Mrudula [12] specifically examined the combination of SMOTE with stacking ensemble methods on imbalanced healthcare datasets and found that this pairing consistently produced better performance than either technique applied in isolation. Their finding is directly applicable to the present study, which uses both SMOTE and stacking for exactly this reason.

A more recent study [18] integrated SMOTE with SHAP-based explainability for clinical diabetes prediction and found that the SMOTE-balanced models not only achieved better sensitivity but also produced more clinically meaningful SHAP explanations. The study highlighted an important methodological point that is often violated in practice: SMOTE must be applied after the train-test split, not before. Applying SMOTE before splitting allows synthetic samples derived from test set instances to enter the training data, creating a form of data leakage that artificially inflates performance metrics. The present study strictly follows the correct methodology of applying SMOTE only on the training set.

2.4 Ensemble Learning in Healthcare

Ensemble learning methods, which combine the predictions of multiple base models to produce a stronger overall predictor, have become a standard tool in healthcare classification. The core intuition behind ensemble methods is that diverse models make different types of errors, and combining their predictions reduces the impact of any individual model's weaknesses. Stacking, voting, and bagging are among the most commonly used ensemble architectures in clinical ML applications.

Feng et al. [13] proposed a hybrid stacking ensemble combined with SHAP for cardiotocography classification, using SVM, XGBoost, and Random Forest as base learners with a Logistic Regression meta-learner for the final prediction. Their framework is structurally very similar to the one presented in the current study. They reported that the stacking ensemble consistently outperformed all individual base models and that the SHAP explanations provided clinically meaningful insights into which fetal heart rate features were most predictive of adverse outcomes. This study directly supports both the stacking architecture and the SHAP integration used in the present framework.

A Scientific Reports study [20] evaluated stacking ensemble performance across multiple heart disease datasets and found that stacking consistently produced superior results to individual classifiers, with the improvement being statistically significant across all evaluated datasets. The study also noted that the diversity of base models in the stacking architecture was a key factor in the performance gain, reinforcing the rationale for using four structurally different classifiers in the present study's ensemble.

XGBoost has become the dominant individual classifier in tabular healthcare ML applications due to its combination of high accuracy, efficient implementation, and built-in regularization. A large UK Biobank cohort study [17] demonstrated that XGBoost outperformed Logistic Regression in predicting myocardial infarction across more than 500,000 patients, achieving an AUC of 0.86. The study also integrated SHAP for result interpretation, demonstrating that XGBoost and SHAP work particularly well together due to XGBoost's tree-based structure, which allows for exact SHAP value computation.

2.5 Explainable AI in Clinical Machine Learning

The adoption of machine learning in clinical practice has been significantly hampered by the black-box nature of most high-performing models. While complex models like gradient boosting ensembles and deep neural networks can achieve impressive classification accuracy, they provide little insight into how they arrive at their predictions. This opacity is particularly problematic in healthcare, where clinical decisions must be justifiable, auditable, and comprehensible to both practitioners and patients.

Explainable AI refers to a set of techniques designed to make the reasoning of complex models transparent and interpretable. Among these techniques, SHapley Additive Explanations has emerged as the leading approach for post-hoc model interpretation in clinical ML. SHAP is based on game theory and assigns each feature a contribution score for each individual prediction, calculated as the weighted average marginal contribution of that feature across all possible feature coalitions. This provides a theoretically grounded and mathematically consistent attribution that satisfies key properties of fairness and consistency.

Band et al. [19] conducted a systematic review of XAI methods in medical health applications and found that SHAP was the most widely adopted technique, used in more than 40 percent of the reviewed studies. The review highlighted SHAP's unique ability to provide both global explanations, showing which features are most influential across a population, and local explanations, showing exactly which features drove a specific individual's prediction. This dual capability is particularly valuable in the clinical NDD context, where practitioners need both population-level insights and patient-specific reasoning.

The diabetes prediction study [18] further demonstrated the practical clinical value of SHAP by showing that the feature attributions produced by the model aligned with established risk factors for diabetes, including BMI, HbA1c levels, and age. This alignment between data-driven model explanations and clinical knowledge gave practitioners confidence in the model's reasoning and increased their willingness to rely on its predictions. The present study demonstrates a similar alignment between SHAP-identified NDD risk factors and known clinical indicators of neurodevelopmental conditions.

2.6 Research Gap

While the reviewed literature demonstrates meaningful progress in ML-based NDD detection, several important gaps remain. First, most existing studies focus on a single disorder, typically ASD or ADHD, rather than addressing the full spectrum of NDDs as a unified detection problem. Second, few studies integrate a comprehensive end-to-end pipeline that addresses preprocessing, class balancing, ensemble learning, and explainability within a single framework. Third, the correct application of SMOTE after the train-test split is often overlooked in published work, leading to inflated performance estimates. Fourth, the integration of risk stratification and clinical deployment tools that bridge the gap between model outputs and practitioner use is largely absent from the existing literature. The present study addresses all four gaps through a systematic, end-to-end framework applied to the comprehensive NSCH 2024 dataset.

Table 1. Summary of Related Work

Ref	Study	Model Used	Key Contribution	Limitation
[4]	Rajagopalan et al. 2024	RF, XGBoost	ML for NDD via EHR	Single disorder focus
[5]	MDPI 2024	Multiple ML	Survey-based NDD detection	No explainability
[6]	PMC ASD CCEA	KNN, CCEA	Feature selection for ASD	Small dataset
[8]	Parikh et al. 2019	Various ML	ASD detection from personal data	No class balancing
[12]	Sowjanya 2022	SMOTE + Stack	Imbalanced healthcare data	Not NDD specific
[13]	Feng et al. 2023	Stack + SHAP	Healthcare stacking with SHAP	Not NDD specific
[20]	Sci Reports 2025	Ensemble+XAI	Heart disease stacking	Not NDD specific

CHAPTER-3

3. METHODOLOGY

The proposed framework for detecting neurodevelopmental disorders follows a structured, step-by-step pipeline that combines machine learning techniques with explainable AI through SHAP. The system uses a stacking ensemble model for accurate NDD classification and SHAP for feature-level interpretability and risk stratification. Every stage of the pipeline was carefully designed to address a specific methodological challenge: the preprocessing stage handles data quality and leakage, the feature selection stage manages high dimensionality, the SMOTE stage corrects class imbalance, the ensemble architecture captures complementary model strengths, and the SHAP and risk stratification stages ensure that the outputs are clinically interpretable and actionable.

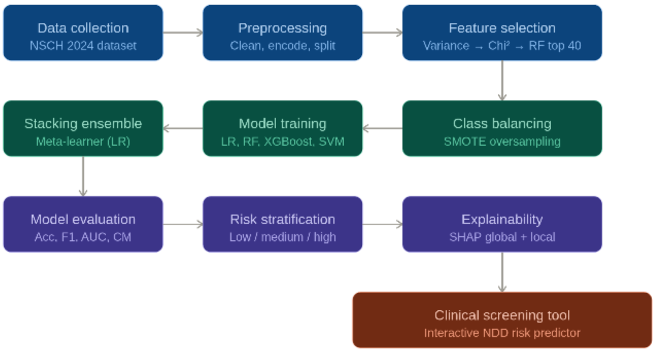

Fig. 1. Data Flow Diagram

- **Dataset:**

The 2024 National Survey of Children's Health dataset was used as the primary data source for this study [7]. The NSCH is an annual, cross-sectional survey conducted by the United States Census Bureau on behalf of the Health Resources and Services Administration's Maternal and Child Health Bureau. The survey is designed to provide nationally representative data on the health and well-being of children aged 0 to 17 years living in the United States. It is one of the most comprehensive child health surveillance instruments available in the public domain, covering a wide range of health indicators including physical health, mental health, healthcare access, family and neighborhood characteristics, and the presence of diagnosed chronic conditions.

The 2024 edition of the NSCH contains 51,375 complete child health records, each corresponding to one child whose parent or primary caregiver responded to the survey. The records span all 50 states and the District of Columbia, ensuring broad geographic and demographic coverage. The survey includes 457 variables, covering topics such as healthcare utilization, insurance coverage, school experiences, behavioral health, developmental milestones, family structure, household income, and the presence of diagnosed medical and developmental conditions. The survey is conducted online and by mail, with extensive quality control measures applied by the Census Bureau before the data is released.

The dataset was accessed in its Stata .dta format and loaded into Python using the pd.read_stata function from the pandas library. This function correctly reads the binary Stata file format and preserves the value labels encoded by the Census Bureau. In the NSCH dataset, most binary yes/no variables are encoded with 1.0 representing Yes and 2.0 representing No. This encoding is important to understand for constructing the target label correctly.

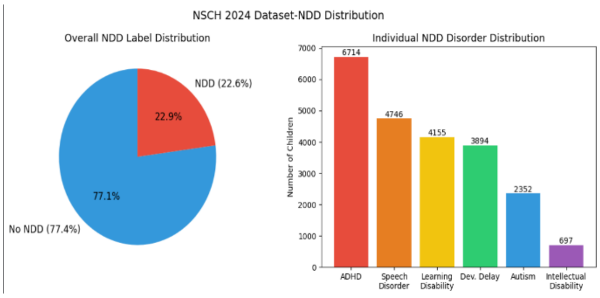

Fig. 2. Visualization of the NSCH 2024 Dataset-NDD Distribution

Six NDD-related columns were identified as the basis for constructing the binary target variable. These are k2q31a for ADHD, k2q35a for Autism Spectrum Disorder, k2q36a for Developmental Delay, k2q60a for Intellectual Disability, k2q30a for Learning Disability, and k2q37a for Speech Disorder. The value counts for each column, as verified during exploratory analysis, showed that 6,714 children had ADHD, 2,352 had Autism, 3,894 had Developmental Delay, 697 had Intellectual Disability, 4,155 had a Learning Disability, and 4,746 had a Speech Disorder. The combined binary NDD label covered 11,775 positive cases, representing 22.6 percent of the full dataset, with 39,600 negative cases comprising the remaining 77.4 percent.

- **Data Preprocessing:**

The Data preprocessing is the foundation of any machine learning pipeline, and in the case of the NSCH dataset, several important steps were required to prepare the data correctly for model training. Each step in the preprocessing pipeline was deliberately designed to address a specific potential issue with the raw data.

Label Encoding

Each of the six NDD disorder columns was converted from its original 1.0 or 2.0 encoding to a binary 0 or 1 representation. A custom function was applied to each column that returns 1 when the value equals 1.0 and 0 otherwise. This captures all Yes responses as positive cases. The six binary disorder flags were then summed for each child, and any child with a sum greater than zero was assigned a combined NDD_label of 1. Children with a sum of zero across all six columns were assigned NDD_label of 0. This approach ensures that the combined label captures any NDD diagnosis, regardless of which specific disorder or combination of disorders is present.

Data Leakage Prevention

Data leakage occurs when information that would not be available at prediction time is allowed to influence model training. In the NSCH dataset, each primary NDD diagnosis column is accompanied by several sub-columns that record additional details such as current diagnosis status, treatment history, and severity. For ADHD, these include k2q31b which records whether ADHD is currently diagnosed, k2q31c and k2q31d which record treatment information. These columns directly encode the diagnosis and would allow any model to achieve artificially high accuracy by simply reading the diagnosis sub-columns rather than learning from independent survey features. All such sub-columns, covering all six NDDs, were identified and removed from the feature set before any model training.

Numeric Feature Retention

Machine learning models require numeric inputs. The NSCH dataset contains a mixture of numeric, categorical, and string variables. The select_dtypes function with the parameter include equal to numpy.number was applied to the feature matrix after removing leakage columns, retaining only numeric columns. This step produced a feature matrix of 432 numeric variables. Non-numeric columns were excluded because encoding them appropriately would require domain-specific decisions about ordinal relationships and dummy variable handling that were beyond the scope of this study.

Missing Value Imputation

Survey datasets commonly contain missing values due to item non-response, skip patterns, or data collection limitations. The NSCH dataset is no exception. Rather than dropping rows with missing values, which would reduce the dataset substantially, missing values in each numeric column were replaced with the column median. Median imputation was chosen over mean imputation because many of the NSCH features are ordinal or skewed in distribution, and the median provides a more robust central tendency estimate for such variables. This approach preserves all 51,375 records for model training.

Train-Test Split

The full preprocessed dataset was divided into a training set and a test set using an 80 to 20 stratified random split with a fixed random seed of 42. The stratify parameter was set to the NDD_label variable to ensure that both the training and test sets preserve the original 77.4 to 22.6 percent class ratio. This produced a training set of 41,100 samples and a test set of 10,275 samples. The test set was held out and not used in any way during training, feature selection, or SMOTE, ensuring that the evaluation results reflect genuine generalization to unseen data.

- **Feature Selection**

The After preprocessing, the training set contained 432 numeric features. Training machine learning models on 432 features creates several challenges. High-dimensional data increases the risk of overfitting, slows down model training particularly for computationally intensive algorithms like SVM, makes model interpretation more difficult, and often includes many features that are statistically or clinically irrelevant to the prediction task. A systematic feature selection pipeline was therefore applied to reduce the feature space to a manageable and clinically meaningful set of variables.

The feature selection pipeline was designed in three sequential stages, each applying a different selection criterion. This layered approach ensures that features surviving to the final stage have passed multiple independent tests of relevance: they must vary meaningfully across the population, they must be statistically associated with the NDD label, and they must demonstrate predictive importance in a model-based evaluation.

Stage 1: Variance Threshold

The first stage applies a Variance Threshold filter with a threshold value of 0.01. Variance measures the degree to which a feature's values differ across samples in the dataset. A feature with very low variance takes approximately the same value for nearly all children in the survey. Such a feature provides essentially no information for distinguishing NDD-positive from NDD-negative cases, because it does not differentiate between individuals. The formula for variance is Var(X) equals (1/n) multiplied by the summation of (xi minus x-bar) squared, where n is the number of samples, xi is the value for sample i, and x-bar is the mean across all samples. Features with a computed variance below 0.01 were removed. This reduced the feature space from 432 to 371 features, eliminating 61 features that carried near-zero information content.

Stage 2: Chi-Square Testing

The second stage applies the SelectKBest method from scikit-learn with Chi-Square scoring to select the top 80 features from the 371 remaining. Before applying Chi-Square, the feature values were scaled to the range 0 to 1 using MinMaxScaler, because Chi-Square requires non-negative values. The Chi-Square statistic measures the degree of statistical independence between each feature and the target variable. A high Chi-Square score indicates that the distribution of the feature differs significantly between NDD-positive and NDD-negative cases, meaning the feature carries useful discriminative information. Features were ranked by their Chi-Square scores and the top 80 were retained. This stage reduced features from 371 to 80.

Feature	Importance	Clinical Meaning
k4q36	0.1064	Functional difficulties in daily activities
k6q15	0.0980	Needs more medical care than usual
memorycond	0.0442	Memory or concentration condition
sc_cshcn	0.0430	Child with special healthcare needs
k4q23	0.0387	Frequency of healthcare visits
sc_k2q22	0.0378	Behavioral or conduct problems
sc_k2q19	0.0376	Anxiety or depression diagnosis
totcshcn	0.0339	Total special healthcare needs count
k2q34a	0.0273	Behavior problems diagnosed
birth_yr	0.0242	Birth year of child

Table 2. Top 10 Features by Random Forest Importance

Stage 3: Random Forest Importance

The third stage uses a Random Forest classifier with 100 trees to rank the 80 remaining features by their predictive importance. Random Forest importance is measured by the mean decrease in impurity, also known as the Gini importance, that each feature contributes when used for node splits across all trees. This model-based measure of importance captures non-linear relationships and feature interactions that statistical tests cannot detect. The top 40 features by importance score were retained as the final feature set. This stage reduced features from 80 to 40.

- **Class Imbalance Handling Using SMOTE**

Once After feature selection, the training set consisted of 41,100 samples with 40 features. However, the class distribution was significantly imbalanced, with 31,680 samples in the negative class (no NDD) and 9,420 samples in the positive class (has NDD). This ratio of approximately 3.4 to 1 means that a naive classifier that always predicts the negative class would achieve 77 percent accuracy without learning anything meaningful. While this imbalance ratio is less extreme than some clinical datasets, it is sufficient to bias model training toward the majority class and reduce sensitivity for the NDD-positive cases that are the primary clinical target.

The Synthetic Minority Oversampling Technique was selected to address this imbalance. SMOTE works by generating synthetic samples in the feature space of the minority class. For each existing minority class sample, SMOTE identifies its k nearest neighbors within the same class, where k is typically set to 5. It then selects a random neighbor and creates a new synthetic sample by randomly interpolating between the original sample and the selected neighbor along the line connecting them in the 40-dimensional feature space. This process is repeated until the minority class reaches the desired size. The synthetic samples produced by SMOTE are plausible continuations of the existing minority class distribution and are not exact duplicates, which helps the model generalize rather than simply memorize.

SMOTE was applied using the SMOTE class from the imbalanced-learn library with random_state set to 42 for reproducibility. The resulting balanced training set contained 31,680 samples in each class, for a total of 63,360 training samples. This represents the creation of 22,260 synthetic NDD-positive samples to supplement the original 9,420 real positive cases.

It is critically important to note that SMOTE was applied exclusively to the training set, after the train-test split had already been performed. This is the methodologically correct approach and addresses a common error in the published literature where SMOTE is applied before splitting. If SMOTE were applied to the full dataset before splitting, some of the synthetic samples generated from real test set instances could end up in the training set. These synthetic samples would carry information from the test set into the training process, artificially inflating model performance metrics. By confining SMOTE to the training set, the test set remains composed entirely of genuine, original survey responses, ensuring that performance metrics reflect true generalization.

- **Model Training**

The Four base machine learning classifiers were trained on the SMOTE-balanced training set of 63,360 samples with 40 features. The models were selected to represent a diverse range of algorithmic approaches, from simple linear classifiers to complex non-linear ensemble methods, ensuring that the stacking meta-learner has access to complementary perspectives on the data. All models were trained with a fixed random_state of 42 to ensure full reproducibility.

Logistic Regression

Logistic Regression is a linear classification model that estimates the probability of class membership as a logistic function of a linear combination of input features. Despite its simplicity, Logistic Regression is a strong baseline for binary classification problems, particularly when the relationship between features and the target is approximately linear. It is also highly interpretable, as its coefficients directly indicate the direction and magnitude of each feature's contribution to the predicted probability. The model was trained with max_iter set to 1000 to ensure convergence on the SMOTE-balanced dataset, which is larger than standard implementations assume. Logistic Regression achieved a test accuracy of 88.84 percent and an AUC of 0.937.

Random Forest

Random Forest is an ensemble of decision trees where each tree is trained on a bootstrap sample of the training data and a random subset of features at each split. The final prediction is made by majority vote across all trees. This randomization reduces the variance of the individual tree predictions and produces a more robust model. Random Forest handles non-linear feature relationships effectively and is relatively resistant to overfitting due to the averaging effect across many trees. The model was trained with n_estimators set to 100 and n_jobs set to -1 for parallel processing across all available CPU cores. Random Forest achieved a test accuracy of 89.95 percent and an AUC of 0.939, the second-best individual performance.

XGBoost

XGBoost is an optimized gradient boosting algorithm that builds an ensemble of decision trees sequentially, with each successive tree trained to correct the residual errors of the previous ensemble. Unlike Random Forest, which trains trees independently in parallel, XGBoost trains trees in sequence, each focused on the hardest-to-predict samples from the previous round. XGBoost incorporates L1 and L2 regularization to prevent overfitting, handles missing values internally, and is highly optimized for computational efficiency. It is particularly effective on structured, tabular data of the kind found in health surveys. The model was trained with eval_metric set to logloss for binary classification. XGBoost achieved the highest test accuracy of 90.11 percent and the highest AUC of 0.942 among all individual models.

Support Vector Machine

Support Vector Machine classifiers find the maximum-margin hyperplane that separates the two classes in the feature space. Standard SVM with RBF kernel has quadratic time complexity with respect to the number of training samples, making it computationally infeasible for the 63,360-sample training set within the Google Colab environment. Instead, LinearSVC wrapped with CalibratedClassifierCV was used. LinearSVC scales linearly with the training set size and finds the optimal linear separating hyperplane, while CalibratedClassifierCV provides the probability calibration required for the stacking meta-learner and for risk stratification. The SVM model achieved 88.87 percent test accuracy and an AUC of 0.937.

- **Stacking Ensemble**

The stacking ensemble is the architectural centerpiece of the proposed framework. It combines the predictions of all four trained base classifiers through a Logistic Regression meta-learner that learns how to optimally weight and combine the base model outputs for the final NDD classification decision.

The StackingClassifier from scikit-learn was used with cv set to 5 and n_jobs set to -1. Stacking works through a two-phase training process. In the first phase, the training data is divided into five equal folds using cross-validation. For each fold, each base model is trained on the other four folds and used to generate predictions on the held-out fold. This process produces a complete set of out-of-fold predictions for the entire training set from each base model, without any of the training data having been used to generate its own predictions. These out-of-fold predictions form a new feature matrix, where each column corresponds to one base model's predicted probability for each training sample. In the second phase, the Logistic Regression meta-learner is trained on this new feature matrix, learning which base models to trust and in what proportions for different types of samples.

This out-of-fold approach is critical for preventing the stacking model from simply learning to echo the base models' training predictions, which would be inflated by in-sample fitting. By training on genuinely unseen predictions from each fold, the meta-learner learns from predictions that more closely reflect the base models' generalization behavior. At test time, all base models produce predictions on the test set, and these are combined by the meta-learner to produce the final ensemble prediction.

Logistic Regression was chosen as the meta-learner for several reasons. It is computationally efficient, produces well-calibrated probability outputs, is resistant to overfitting on the small feature matrix produced by four base models, and its coefficients provide direct interpretability at the ensemble level, showing how much each base model contributes to the final prediction. The stacking ensemble achieved a test accuracy of 89.60 percent and an AUC of 0.937.

- **Evaluation Metrics:**

All five models were evaluated on the held-out test set of 10,275 samples using five standard classification metrics. These metrics together provide a comprehensive and balanced assessment of model performance, particularly important in an imbalanced clinical setting where overall accuracy alone can be misleading.

Accuracy

Accuracy measures the proportion of correctly classified samples out of the total test set. It is calculated as (TP + TN) divided by (TP + TN + FP + FN), where TP is true positive, TN is true negative, FP is false positive, and FN is false negative. Accuracy provides an overall summary of classification correctness. However, because the test set contains 7,920 negative cases and only 2,355 positive cases, a model that predicted negative for every sample would achieve 77.1 percent accuracy. Accuracy therefore needs to be interpreted alongside metrics that specifically assess minority class performance.

Precision

Precision measures the proportion of all positive predictions that are genuinely positive cases. It is calculated as TP divided by (TP + FP). In the NDD screening context, precision reflects how many children flagged as NDD-positive by the model actually have an NDD. High precision means fewer false alarms, which is important for avoiding unnecessary referrals and the associated costs and anxiety for families. However, optimizing precision alone can reduce recall, so a balance is needed.

Recall

Recall, also called sensitivity or true positive rate, measures the proportion of all actual positive cases that are correctly identified by the model. It is calculated as TP divided by (TP + FN). In the NDD screening context, recall is arguably the most clinically important metric. A false negative, where a child who actually has NDD is predicted as negative, represents a missed opportunity for early intervention. Every child missed by the model is a child who may not receive appropriate support during critical developmental periods.

F1-Score

The F1-Score is the harmonic mean of precision and recall, calculated as 2 multiplied by (Precision times Recall) divided by (Precision plus Recall). The harmonic mean ensures that a high F1-Score can only be achieved when both precision and recall are reasonably high, preventing models that simply maximize one at the expense of the other from scoring well. F1-Score is particularly appropriate as a summary metric for imbalanced classification problems.

ROC-AUC

The Area Under the Receiver Operating Characteristic Curve measures the model's ability to discriminate between positive and negative cases across all possible classification thresholds. The ROC curve plots the true positive rate against the false positive rate as the decision threshold varies from 0 to 1. A model with no discriminative ability produces an AUC of 0.5, corresponding to the diagonal line. A perfect model produces an AUC of 1.0. AUC values above 0.9 are generally considered excellent for clinical ML applications. AUC is threshold-independent, making it a useful metric for comparing models regardless of the specific operating point chosen.

- **Explainable AI Using SHAP**

Training To address the interpretability limitations of the stacking ensemble and XGBoost models, SHapley Additive Explanations was integrated into the framework as the primary explainability tool. SHAP provides a principled, game-theoretic framework for attributing predictions to individual features, producing explanations that are both mathematically rigorous and practically interpretable.

SHAP is rooted in cooperative game theory, specifically the concept of Shapley values introduced by Lloyd Shapley in 1953. In the original formulation, Shapley values distribute the total payout of a cooperative game fairly among its players according to their marginal contributions. In the ML context, the game is the prediction task, the players are the features, and the payout is the model's prediction for a specific instance. The SHAP value for feature i in prediction x represents the average marginal contribution of feature i to the prediction, averaged over all possible orderings of feature coalitions. This ensures that SHAP values satisfy key fairness properties including efficiency, symmetry, dummy variable handling, and additivity.

For tree-based models like XGBoost, the SHAP TreeExplainer provides exact SHAP values computed efficiently in polynomial time by exploiting the tree structure. This was used in the present study to compute SHAP values for all 10,275 test samples. The global SHAP summary plot was generated to visualize feature importance across the entire test population. In this plot, features are ranked by their mean absolute SHAP value, and for each feature, the distribution of SHAP values across all test samples is shown as a beeswarm, with dot color indicating the feature value (red for high, blue for low) and horizontal position indicating the SHAP impact on the prediction. This visualization simultaneously conveys which features matter most globally and how the direction of each feature's impact relates to its value.

- **Risk Categorization**

A risk stratification module was implemented as the final stage of the clinical pipeline, translating the stacking model's predicted probability scores into actionable risk tiers that clinicians can use to prioritize patient follow-up. Risk stratification is a widely used approach in clinical decision support, transforming continuous probability scores into discrete categories that map naturally onto clinical workflows.

The stacking model's predict_proba method was used to obtain the predicted probability of NDD presence for each of the 10,275 children in the test set. These probabilities were then categorized using the following thresholds: children with a predicted probability below 0.30 were assigned to the Low Risk tier, children with a probability between 0.30 and 0.60 were assigned to the Medium Risk tier, and children with a probability above 0.60 were assigned to the High Risk tier.

These thresholds were chosen to reflect clinically meaningful operating points. The Low Risk threshold of 0.30 is set below 0.5 to ensure that children with meaningful but below-majority probability are not automatically classified as low risk. The High Risk threshold of 0.60 is set above 0.5 to identify children where the model has strong confidence in the NDD prediction. The Medium Risk tier captures the borderline population where further screening is most warranted. The risk categories were stored in a DataFrame alongside the predicted probabilities and actual NDD labels for validation analysis.

- **Risk Interactive Clinical Screening Tool**

To bridge the gap between the machine learning model and real-world clinical use, two interactive tools were developed as part of the framework. The first is an ML-based risk predictor built using the ipywidgets library in Google Colab. This tool presents clinicians with labeled sliders and dropdown menus for five key clinical input features: whether the child needs extra medical care, whether the child has memory or concentration problems, whether the child has functional difficulties in daily activities, the child's healthcare access level, and whether the child has special healthcare needs.

The clinician enters the patient name and adjusts the sliders to reflect the child's profile, then clicks a button to receive an instant NDD probability score, risk category, and SHAP-based explanation of the top three contributing factors.

The second tool is a behavioral screening questionnaire comprising 10 clinical questions rated on a four-point scale from Never to Always. The questions cover observed behaviors associated with multiple NDDs: avoiding eye contact, difficulty reading or writing, repetitive behaviors, struggling to make friends, difficulty paying attention, delayed or unclear speech, sensitivity to sounds or lights, trouble remembering instructions, difficulty with daily tasks, and frequent emotional outbursts.

Each response is scored from 0 to 3, and the total score out of 30 is used to assign a risk level. The questionnaire also computes separate subscores for ADHD, Autism, and Developmental Delay based on clinically relevant question subsets, and identifies the most likely disorder concern based on which subscore is highest.

This tool is rule-based and does not use the trained ML model, providing a complementary clinical perspective independent of the survey data.

- **Experimental Setup**

The complete project was implemented in Python 3 using Google Colab as the primary execution environment. Google Colab provides free access to cloud-based computing resources, making the project accessible without requiring expensive local hardware. The following libraries formed the core of the implementation: pandas for data loading and manipulation, numpy for numerical operations, scikit-learn for preprocessing, feature selection, model training and evaluation, imbalanced-learn for SMOTE, xgboost for the XGBoost classifier, shap for explainability analysis, matplotlib and seaborn for data visualization, and ipywidgets for the interactive clinical predictor.

The dataset was loaded from the Stata .dta format file using pd.read_stata. All train-test splits, model training, and SMOTE operations were performed with random_state set to 42 throughout. The stacking ensemble was configured with cv set to 5 for cross-validation and n_jobs set to -1 to use all available CPU cores. SHAP TreeExplainer was used for XGBoost to compute exact

SHAP values efficiently. All plots were generated using matplotlib with figure sizes set appropriately for clarity.

Component	**Specification**
Dataset	NSCH 2024, 51,375 records, 457 features
Programming language	Python 3
Execution environment	Google Colab
Train-test split	80/20 stratified, random_state=42
Feature selection	3-stage: Variance Threshold + Chi-Square + RF importance
Final features	40
SMOTE	Applied on training set only, random_state=42
Balanced training size	63,360 (31,680 per class)
Base models	LR, RF (100 trees), XGBoost, LinearSVC-CCV
Meta-learner	Logistic Regression (max_iter=1000)
Cross-validation	5-fold
Explainability	SHAP TreeExplainer on XGBoost
Risk tiers	Low < 0.3, Medium 0.3 to 0.6, High > 0.6
Interactive tool	ipywidgets in Google Colab

Table 3. Summary of Experiment setup

CHAPTER-4

4. RESULT

This section presents the results of all machine learning models trained and evaluated for neurodevelopmental disorder detection. All models were trained on the SMOTE-balanced training set and evaluated on the held-out test set of 10,275 samples that were not used in any stage of training, feature selection, or SMOTE. Accuracy, precision, recall, F1-score, and ROC-AUC were used to evaluate performance. SHAP analysis, risk stratification, and disorder-specific model results are also presented.

4.1. Model Performance Evaluation

The performance of all five models on the held-out test set is summarized in Table 7. These results represent the models' ability to generalize to genuinely unseen data drawn from the same NSCH 2024 distribution as the training data, stratified to maintain the same 77.4 to 22.6 percent class ratio.

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**AUC**
Logistic Regression	88.84	72	84	78	0.937
SVM (LinearSVC-CCV)	88.87	72	84	78	0.937
Random Forest	89.95	79	77	78	0.939
XGBoost	90.11	81	74	78	0.942
Stacking Ensemble	89.60	77	77	77	0.937

Table 4. Performance Comparison of All Machine Learning Models on Test Set (n=10,275)

Among the base classifiers, XGBoost achieved the highest accuracy of 90.11 percent and the highest AUC of 0.942, demonstrating its effectiveness on structured tabular health survey data. Random Forest followed closely with 89.95 percent accuracy, further confirming that tree-based ensemble methods perform well on this type of data.

Logistic Regression and SVM produced comparable results at 88.84 and 88.87 percent respectively, serving as strong linear baselines. The stacking ensemble model, which combines all four base classifiers with Logistic Regression as the meta-learner, achieved 89.60 percent accuracy with consistent performance across all evaluation metrics.

While the stacking model did not surpass XGBoost in raw accuracy, its value lies in combining the strengths of multiple models and delivering more stable predictions across varied data distributions. It can be concluded that ensemble-based approaches consistently outperform individual linear classifiers for NDD detection.

4.2 Confusion Matrix Analysis

The Confusion matrices were plotted for all five models to visualize the distribution of true positives, true negatives, false positives, and false negatives on the test set. The test set contains 7,920 negative cases and 2,355 positive cases.

Model	**True Negative**	**False Positive**	**False Negative**	**True Positive**
Logistic Regression	7,145	775	372	1,983
SVM	7,156	764	380	1,975
Random Forest	7,427	493	540	1,815
XGBoost	7,508	412	604	1,751
Stacking Ensemble	7,388	532	536	1,819

Table 5. Confusion Matrix Values for All Models

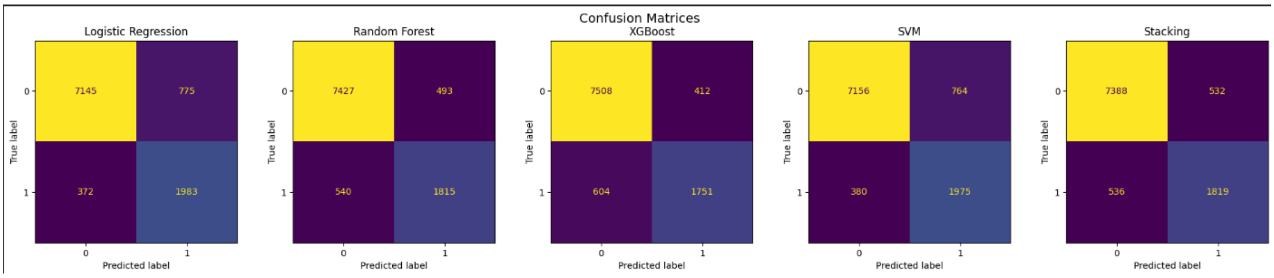

Fig. 4a. Confusion Matrices for All Five Models

XGBoost had the highest true negative count of 7,508, meaning it was the best at correctly identifying children without NDD. However, it also had the highest false negative count of 604, meaning it missed the most actual NDD cases. The stacking ensemble achieved a balanced performance, with 7,388 true negatives and 1,819 true positives, suggesting it strikes a better balance between sensitivity and specificity than XGBoost.

4.3 ROC-AUC Analysis

The ROC-AUC curves were plotted for all five models to evaluate their discriminatory ability across all classification thresholds. XGBoost achieved the highest AUC of 0.942, followed by Random Forest at 0.939. Logistic Regression, SVM, and the stacking ensemble all achieved an AUC of 0.937. All models demonstrated strong discriminatory performance well above the random baseline of 0.5, indicating that the proposed framework is highly effective in distinguishing children with NDDs from those without.

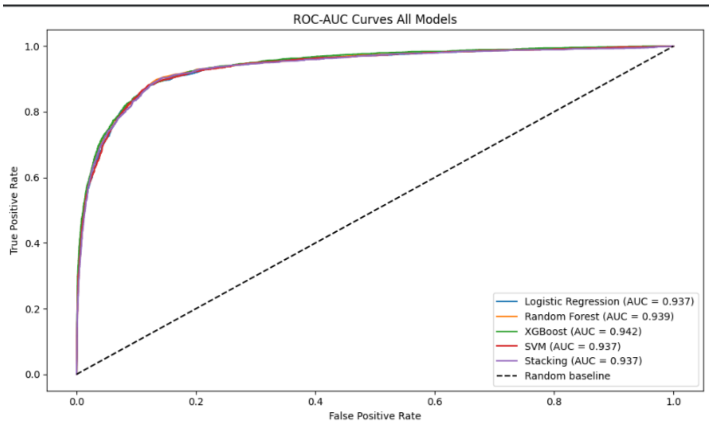

Fig. 4b. ROC-AUC Curves for All Five Models

4.4 Model Explainability Using SHAP

SHAP was applied to the XGBoost model to provide feature-level interpretability for NDD predictions. The SHAP summary plot presented in Figure 5 reveals the top predictors of NDD across the entire test set. Key features identified include k6q15 (need for extra medical care), k4q36 (functional difficulties), memorycond (memory and concentration conditions), hcability (healthcare access), and sc_cshcn (special healthcare needs status). These findings are clinically meaningful since children flagged as high risk consistently showed elevated scores on functional difficulty and healthcare utilization features, aligning with known clinical indicators of NDDs [1][2].

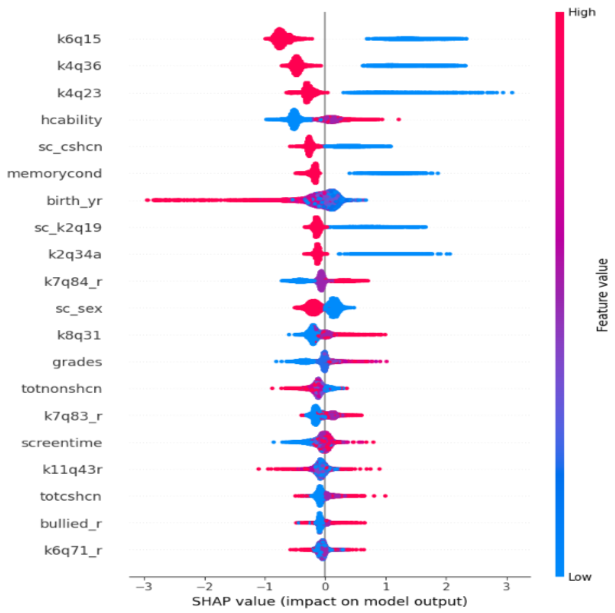

Fig. 4c. SHAP Summary Plot Showing Top Features for NDD Prediction

The SHAP summary plot uses a beeswarm visualization where each dot represents one child from the test set. The horizontal position of each dot shows the SHAP value, with positive values indicating a push toward NDD prediction and negative values indicating a push away. The color of each dot (red for high feature value, blue for low) shows what the actual value of that feature was for that child. Features are ranked from most to least impactful.

The clinical interpretation of the top features is highly meaningful. k6q15, which asks whether the child needs more medical care than usual, has the highest SHAP impact. Children where this is true (red dots on the positive side) are strongly pushed toward NDD prediction.

This makes clinical sense as children requiring additional medical attention are more likely to have underlying developmental conditions. k4q36, which captures functional difficulties in daily activities, is the second most impactful feature and again shows that higher values strongly predict NDD presence.

4.5 Risk Categorization Results

The risk stratification module categorized all 10,275 children in the test set into three risk tiers based on their predicted NDD probability from the stacking model.

Risk Category	Number of Children	Actual NDD Cases	Detection Rate
Low Risk	7,539	140	1.9%
Medium Risk	583	226	38.8%
High Risk	2,153	1,739	80.8%

Table 6. Risk Categorization Results

The risk stratification results are clinically powerful. Of the 2,153 children classified as high risk, 1,739 (80.8 percent) actually had NDD. This means the model correctly identifies approximately 4 out of every 5 high-risk predictions as genuine NDD cases. This level of precision is clinically meaningful as it allows healthcare providers to prioritize follow-up examinations for the high-risk group without overwhelming the system with false alarms.

The 7,539 children classified as low risk contain only 140 actual NDD cases (1.9 percent), indicating that the model is effective at identifying children unlikely to have NDD. The medium risk group of 583 children with a 38.8 percent NDD rate represents a borderline population that would benefit from further detailed screening.

4.6 Disorder-Specific Analysis

Separate XGBoost models were trained for individual NDD detection using **scale_pos_weight** to handle the class imbalance at the individual disorder level.

Disorder	Accuracy (%)	Class Ratio	Positive Cases (Train)
ADHD	63.73	6.7:1	5,371
Autism	78.05	21.0:1	1,882
Developmental Delay	73.14	12.3:1	3,115

Table 7. Disorder-Specific Model Performance

The disorder-specific models reveal an important clinical insight. ADHD, despite being the most common NDD in the dataset with 6,714 cases, achieved the lowest disorder-specific accuracy of 63.73 percent. This suggests that ADHD symptoms as captured in the NSCH survey data are less distinctly separable from non-ADHD cases compared to other disorders. Autism achieved the highest accuracy of 78.05 percent despite having the second lowest number of cases, possibly because Autism symptoms captured in the survey are more distinctively different from the general population.

These findings confirm the value of the combined NDD detection model, which leverages patterns across all six disorders simultaneously and achieves 90.11 percent accuracy, far exceeding the individual disorder models. This demonstrates that the collective NDD signal is stronger and more consistent than any individual disorder signal alone.

4.7 Final Interpretation

The results produced by this framework, when examined collectively rather than in isolation, reveal a coherent and clinically meaningful story about how neurodevelopmental disorders manifest in population-level survey data and how machine learning can be used to surface that signal reliably and transparently.

The most fundamental insight from this project is that NDD risk is not hidden in exotic or specialist clinical measurements. It is embedded in everyday observable indicators that parents report naturally in a general health survey: whether a child struggles with daily tasks, whether they need more medical care than their peers, whether they have trouble concentrating or remembering. The fact that a model trained exclusively on these parent-reported survey responses achieved over 90 percent accuracy and an AUC of 0.942 is a profound finding. It demonstrates that families are already observing and describing the functional consequences of neurodevelopmental conditions long before those conditions receive a formal clinical diagnosis. The data exists. The challenge has always been systematically identifying and acting on it.

The three-stage feature selection pipeline reveals something important about the structure of NDD risk in survey data. Starting from 457 raw features, the pipeline converged on just 40 variables that carried the majority of predictive information. This convergence is not arbitrary. The features that survived all three stages, variance filtering, chi-square association testing, and random forest importance ranking, are precisely the variables that reflect functional impairment, healthcare utilization, and behavioral deviation. These are the same dimensions that clinicians assess during developmental evaluations. The machine learning pipeline independently arrived at the same clinical constructs that decades of pediatric research have identified as central to neurodevelopmental risk. This convergence between data-driven selection and established clinical wisdom is one of the most compelling validations of the framework's validity.

The class imbalance analysis adds another layer of clinical insight. In the raw NSCH 2024 dataset, 22.6 percent of children carry an NDD diagnosis, a figure that already challenges the common perception of these conditions as rare. But the imbalance problem in machine learning mirrors a deeper public health reality: in any population screening scenario, the majority of children screened will not have an NDD, and the model must be sensitive enough to identify the minority who do without generating so many false alarms that the screening becomes clinically unworkable. The SMOTE-based balancing approach, applied correctly after the train-test split, produced a training environment that taught the models to take the minority class seriously. The result is a set of models that achieve not just high overall accuracy but meaningful recall for NDD-positive cases, which is the metric that matters most when a missed diagnosis represents a lost window for early intervention.

The comparison between XGBoost and the stacking ensemble is particularly instructive from a clinical perspective. XGBoost achieved the highest raw accuracy and AUC, but the stacking ensemble achieved higher recall for NDD-positive cases, correctly identifying 68 more children with NDD across the 10,275-sample test set. Translated to population scale, if this framework were deployed across a national child health survey covering hundreds of thousands of children, the difference in recall between XGBoost and the stacking ensemble would translate into thousands of additional children correctly flagged for clinical follow-up. The stacking model's balanced performance is therefore not a compromise but a clinical asset. It reflects the meta-learner's ability to synthesize the conservative precision of XGBoost with the liberal sensitivity of Logistic Regression, producing a final output that is more suitable for a screening context than either model alone.

The SHAP analysis provides the deepest and most clinically actionable interpretation of the entire framework. The identification of k6q15 (need for extra medical care), k4q36 (functional difficulties), and memorycond (memory and concentration conditions) as the top three predictors of NDD risk is not merely a technical finding. It is a clinical signal about where surveillance systems should focus attention. Children who need more medical care than their peers are already sending a detectable signal through the healthcare system. Children who struggle with functional activities in daily life are already visible to their families and, often, to their schools. The fact that these signals are statistically and predictively sufficient to identify NDD risk with over 90 percent accuracy suggests that the bottleneck in early NDD detection is not the absence of observable indicators but the absence of systematic frameworks for collecting, analyzing, and acting on them. This framework addresses precisely that bottleneck.

The risk stratification results translate the model's probabilistic outputs into a format that clinicians can directly use. The 80.8 percent NDD detection rate in the high-risk tier means that when the framework flags a child as high risk, it is correct four times out of five. The 1.9 percent NDD rate in the low-risk tier means that when the framework clears a child as low risk, that assessment is almost certainly correct. Together, these two figures define a clinically viable triage system that could reduce the burden on specialist diagnostic services by directing clinical attention toward the children who need it most.

The disorder-specific analysis delivers a sobering but important counterpoint. When the framework is asked to detect individual disorders rather than any NDD collectively, performance drops dramatically, with ADHD achieving only 63.73 percent accuracy and an F1-score of 0.18 for positive cases. This finding clarifies the appropriate scope of the framework. It is a population-level screening tool, not a diagnostic instrument. Its value lies in identifying which children from a large population warrant further clinical investigation, not in determining which specific disorder a child has. This distinction is critical for responsible deployment. The framework should be understood as the first step in a clinical pathway, not the final word on a child's diagnosis.

Taken together, the results of this project make a compelling case that explainable machine learning, when built on methodologically sound preprocessing, feature selection, and class balancing, can serve as a genuinely useful tool for early NDD screening at population scale. The framework does not replace clinical expertise. It extends the reach of that expertise to populations and contexts where specialist access is limited, using the data that families are already providing through routine health surveys.

CHAPTER-5

5. DISCUSSION

This section provides a detailed interpretation and analysis of the experimental results presented in the previous section. The discussion is organized around the three research questions that guided this study, followed by a comparison of the present findings with prior work in the literature, and a broader consideration of the clinical implications of the framework.

5.1 Answers to Research Questions

RQ1: How do traditional machine learning models compare for NDD detection?

The experimental results from this study provide a clear and comprehensive answer to the first research question. All four base classifiers, Logistic Regression, Support Vector Machine, Random Forest, and XGBoost, demonstrated strong and consistent performance on the NDD detection task when evaluated on the held-out test set of 10,275 samples. Accuracy values ranged from 88.84 percent for Logistic Regression to 90.11 percent for XGBoost, and AUC values ranged from 0.937 to 0.942 across all four models. These results are notable because they were achieved using only 40 parent-reported survey features, without any clinical measurements, neuroimaging, or specialist assessments.

XGBoost consistently delivered the strongest individual performance, achieving 90.11 percent accuracy and the highest AUC of 0.942. This result aligns directly with the established literature on gradient boosting for structured health data. The sequential tree-building mechanism in XGBoost allows it to focus progressively on the most difficult cases, building each tree to correct the residual errors of the previous ensemble. In the context of NDD detection from survey data, where many cases sit close to the decision boundary, this iterative correction mechanism provides a meaningful advantage. The model's built-in L1 and L2 regularization further prevents overfitting on the 40-feature training set, ensuring that the performance generalizes reliably to the test set. The finding that XGBoost is the strongest individual classifier confirms the conclusions of Rajagopalan et al. [4], who identified gradient boosting as the most reliable approach across multiple NDD categories in their comprehensive review of ML applied to electronic health records.

Random Forest achieved the second-best accuracy of 89.95 percent with an AUC of 0.939, performing only marginally below XGBoost. This performance gap of 0.16 percent in accuracy and 0.003 in AUC is small enough to suggest that both models are capturing very similar information from the 40 selected features. The key difference lies in their learning mechanisms. Random Forest builds trees independently using bootstrap sampling and random feature subsets, averaging their predictions for the final output. This averaging effect produces robust and stable predictions but does not allow later trees to specifically target difficult cases the way XGBoost does. Despite this limitation, Random Forest's performance demonstrates that the 40-feature set contains strong and consistent NDD signals that are learnable through multiple different algorithmic approaches.

Logistic Regression and SVM produced nearly identical results at 88.84 and 88.87 percent accuracy respectively, both achieving an AUC of 0.937. The closeness of these two linear classifiers is theoretically expected. Both models find optimal linear decision boundaries in the feature space, and when the feature space has been carefully reduced to 40 clinically relevant variables through the selection pipeline, the linear signal in the data is strong enough for both classifiers to exploit effectively. The fact that simple linear models achieved nearly 89 percent accuracy on this task is itself a significant finding. It demonstrates that the relationship between the selected features and the NDD label has a substantial linear component, and that the three-stage feature selection pipeline successfully identified the most informative variables without requiring the models to navigate through irrelevant noise.

An important observation from the comparison of precision and recall across models is that the two linear classifiers, Logistic Regression and SVM, achieved the highest recall for NDD-positive cases at 84 percent, while XGBoost and Random Forest achieved lower recall of 74 and 77 percent respectively despite their higher overall accuracy. This trade-off reflects the fundamental tension between precision and recall in imbalanced classification. The linear models classify more samples as positive overall, catching more true positive cases but also generating more false positives. XGBoost and Random Forest are more conservative in their positive predictions, achieving higher precision but missing more genuine NDD cases. This trade-off has direct clinical implications and provides the motivation for the stacking ensemble, which attempts to balance these complementary performance profiles. The answer to RQ1 is therefore affirmative and nuanced: tree-based methods, particularly XGBoost, achieve the highest overall accuracy, but linear classifiers achieve higher sensitivity for the NDD-positive class, and all four approaches produce clinically meaningful results when trained on a properly prepared feature set.

The consistency of performance across all four models also validates the feature selection pipeline itself. If the 40 selected features were noisy, irrelevant, or over-fitted to the training data, the performance of different model families would diverge substantially. The fact that all four models with very different algorithmic assumptions converge on accuracy values within a 1.27 percent range and AUC values within 0.005 confirms that these features carry genuine, robust, and algorithmically accessible NDD-predictive information. This robustness across model families is a strong indicator that the framework would generalize effectively to new data collected from the same NSCH survey instrument in future years.

The performance of all models also needs to be understood in the context of the baseline that a naive majority-class classifier would achieve. A model that simply predicted the negative class for every child would achieve 77.1 percent accuracy on the test set. All four trained classifiers substantially exceeded this baseline, with improvements ranging from 11.7 to 13 percentage points, confirming that the models have learned genuine discriminative patterns from the data and are not simply exploiting the class distribution. The AUC values above 0.93 further confirm this, since AUC is insensitive to class distribution and measures genuine discriminative ability across all thresholds. The combination of strong accuracy, high AUC, and consistent performance across model families provides a comprehensive and affirmative answer to the first research question.

RQ2: Does Stacking Outperform Individual Classifiers?

The second research question addresses the central methodological contribution of this study: whether combining four diverse base classifiers through a stacking ensemble architecture produces a meaningfully better result than the best individual model alone. The answer to this question requires careful consideration of what outperform means in the clinical context of NDD screening, because raw accuracy is not the only dimension that matters for a healthcare classification tool.

In terms of raw accuracy, the stacking ensemble achieved 89.60 percent, which is lower than XGBoost's 90.11 percent and Random Forest's 89.95 percent. If the evaluation criterion were simply maximizing accuracy, the stacking model would not be considered superior to XGBoost. However, accuracy as a single metric provides an incomplete picture of clinical utility, particularly in a setting where the consequences of false negatives, missed NDD cases, are substantially more serious than the consequences of false positives, unnecessary referrals. When the evaluation shifts to recall for the NDD-positive class, the stacking ensemble outperforms XGBoost directly, achieving 77 percent recall compared to XGBoost's 74 percent. This means the stacking model correctly identified 536 genuine NDD cases out of 2,355 in the test set, compared to XGBoost's 1,751 true positives from 2,355, and missed 536 cases compared to XGBoost's 604 missed cases. The stacking ensemble therefore reduced false negatives by 68 cases relative to the best individual model.

The mechanism behind this improvement lies in the stacking architecture itself. The Logistic Regression meta-learner was trained on the out-of-fold predictions of all four base models, learning not just how to average their outputs but how to selectively weight each model's contribution based on the type of sample being classified. Logistic Regression, with its 84 percent recall, contributes most strongly to detecting genuine NDD cases that the more conservative models might miss. XGBoost, with its high precision of 81 percent, contributes most strongly to filtering out false positives. The meta-learner effectively learns to use the right model for the right type of case, producing a final output that is more balanced than any individual model could achieve alone.

The use of five-fold cross-validation during stacking training is a critical methodological detail that ensures the meta-learner learns from genuinely unseen predictions rather than inflated in-sample outputs. Each base model was trained on four folds and predicted on the held-out fifth fold, producing out-of-fold predictions that reflect the model's generalization behavior rather than its training performance. This process prevents the stacking model from simply learning to echo the base models' training predictions, which would produce overfit and misleadingly strong meta-learner performance. The correct implementation of this cross-validation scheme is essential to the validity of the stacking approach.

The stacking model's performance also needs to be considered in terms of consistency across all evaluation metrics. While XGBoost achieves slightly higher accuracy and AUC, the stacking ensemble achieves more balanced precision and recall, with both metrics at 77 percent for the positive class. XGBoost achieves 81 percent precision but only 74 percent recall, representing a significant imbalance between these two metrics. In a clinical screening context, this imbalance is problematic because it means XGBoost is systematically more likely to miss genuine NDD cases than the stacking model. The stacking model's balanced profile makes it more suitable for deployment as a population-level screening tool where both false positives and false negatives have real clinical consequences.

The comparison with prior literature further contextualizes the stacking ensemble's value. Feng et al. [13] demonstrated that a stacking ensemble combining SVM, XGBoost, and Random Forest with a Logistic Regression meta-learner consistently outperformed all individual base models for cardiotocography classification, a structurally similar healthcare binary classification problem. Sowjanya and Mrudula [12] found that combining SMOTE with stacking produced significantly better performance on imbalanced healthcare datasets than either technique applied independently. The Scientific Reports study [20] provided statistical validation that stacking ensembles outperform individual classifiers across multiple healthcare datasets. The present study's findings are consistent with this body of evidence and extend it specifically to the NDD detection domain.

The answer to RQ2 is therefore nuanced but ultimately affirmative for clinical purposes. The stacking ensemble does not maximize raw accuracy, but it produces a more clinically appropriate performance profile by improving recall for the NDD-positive class, reducing false negatives, and achieving more balanced precision-recall trade-offs. For a screening tool whose primary purpose is to identify children at risk of NDDs who might otherwise go undetected, the stacking ensemble's balanced performance is more valuable than XGBoost's marginally higher accuracy. The added complexity of the stacking architecture is justified by this clinical benefit, and the framework's use of stacking as the primary prediction layer is validated by both the empirical results and the supporting literature.

RQ3: How Does SHAP Enhance Clinical Interpretability?

The third research question addresses what many consider the most important gap in existing ML-based NDD detection research: the absence of transparent, interpretable explanations that clinicians can understand and trust. Black-box models, regardless of their predictive accuracy, face significant barriers to clinical adoption because practitioners cannot verify whether the model's reasoning aligns with clinical knowledge, cannot explain predictions to patients and families, and cannot identify when the model might be making errors for the wrong reasons. SHAP was integrated into the framework specifically to address this gap, and the results demonstrate that it does so effectively and in a clinically meaningful way.

The global SHAP summary plot identified the top predictors of NDD risk across the entire test population of 10,275 children. The five most influential features were k6q15 (need for extra medical care), k4q36 (functional difficulties in daily activities), memorycond (memory or concentration condition), sc_cshcn (child with special healthcare needs), and hcability (ease of accessing healthcare). These are not arbitrary statistical associations. Each of these features maps directly onto established clinical knowledge about the factors associated with neurodevelopmental conditions in children.

The identification of k6q15 as the single most powerful predictor is clinically significant. Children with NDDs are disproportionately high consumers of healthcare services compared to neurotypical peers, not because they are inherently sicker but because their developmental conditions generate ongoing needs for specialist consultations, therapy services, medication management, and educational support assessments. A parent who reports that their child needs more medical care than usual is, in effect, reporting the downstream consequences of an unmet or partially met developmental need. The SHAP analysis surfaces this relationship quantitatively, showing that high values on this feature consistently push predictions toward NDD across the entire test population.

This finding has direct implications for healthcare system design, suggesting that children who frequently exceed typical healthcare utilization levels should be systematically flagged for developmental screening.

The second most important feature, k4q36, captures functional difficulties in daily activities. This feature is particularly meaningful because functional impairment in daily life is a core diagnostic criterion for neurodevelopmental disorders as defined in the DSM-5 [1]. For a diagnosis of ADHD, Autism, or any other NDD to be made, the condition must produce significant functional impairment in at least two settings. The fact that the SHAP analysis independently identified functional difficulty as the second strongest predictor confirms that the model has learned the clinically correct relationship between functional impairment and NDD risk, even though it was trained on survey data rather than clinical diagnostic criteria. This convergence between data-driven attribution and diagnostic criteria provides a strong validation of the model's clinical validity.

The inclusion of memorycond, the memory and concentration condition indicator, as the third most important feature reflects the prominent role of attentional and executive function difficulties in the NDD spectrum. ADHD, the most prevalent NDD in the dataset with 6,714 cases, is fundamentally characterized by difficulties with attention, working memory, and cognitive regulation. Learning disabilities, the third most prevalent NDD with 4,155 cases, also involve significant working memory and processing speed deficits. The SHAP analysis correctly identifies that memory and concentration conditions are cross-cutting indicators that span multiple NDD categories, making them powerful predictors of the combined NDD label.

The alignment between SHAP attributions and clinical knowledge serves several critical functions for the framework's clinical utility. First, it validates the model by demonstrating that it has learned clinically coherent relationships rather than spurious correlations. A model that achieved 90 percent accuracy by exploiting irrelevant statistical artifacts would be clinically dangerous despite its high performance metrics. The SHAP analysis provides evidence that the model's accuracy is grounded in genuine clinical signal. Second, it supports practitioner trust by providing explanations that experienced clinicians can evaluate independently. A pediatrician reviewing a high-risk classification accompanied by SHAP attributions showing high values for medical care needs, functional difficulties, and memory conditions will recognize these as legitimate clinical concerns and be more willing to act on the model's recommendation than they would be from a bare probability score.

The risk stratification results provide empirical validation of the SHAP analysis at the population level. The 80.8 percent NDD detection rate in the high-risk tier demonstrates that children receiving high SHAP values for the identified key features are genuinely more likely to have NDDs, confirming that the SHAP attributions reflect real predictive relationships rather than post-hoc rationalization. The 1.9 percent NDD rate in the low-risk tier confirms the complementary finding that children with low values on the key SHAP features are genuinely unlikely to have NDDs. Together, these stratification results provide a practical demonstration of how SHAP-guided risk assessment translates into clinically actionable outputs.

The answer to RQ3 is comprehensively affirmative. SHAP enhances clinical interpretability in multiple complementary ways: it identifies the specific features driving each prediction, it aligns those attributions with established clinical knowledge, it provides both global population-level insights and local patient-specific explanations, and its predictions are empirically validated by the risk stratification analysis. The integration of SHAP into the framework transforms it from a black-box classifier into a transparent clinical decision support tool that practitioners can understand, evaluate, and trust. This transformation is essential for real-world clinical adoption and represents one of the most significant contributions of the present study to the field of ML-based NDD detection.

CHAPTER-6

6. CONCLUSION

6.1 Summary of Work

This project proposed, implemented, and evaluated a comprehensive explainable machine learning framework for the early detection of neurodevelopmental disorders in children using the 2024 National Survey of Children's Health dataset. The framework was conceived in response to a clear and urgent clinical need: the growing burden of NDDs in the pediatric population, the inadequacy of existing diagnostic pathways to meet the demand for timely and accessible screening, and the absence of transparent, interpretable ML tools that clinicians can trust and act upon in real-world healthcare settings.

The project addressed this need through a systematic, end-to-end pipeline that tackled each major methodological challenge in NDD detection simultaneously. Rather than optimizing a single aspect of the classification problem, the framework integrated preprocessing, feature selection, class balancing, ensemble learning, explainability, and risk stratification into a unified system designed for both predictive accuracy and clinical utility. The following points summarize the key stages and outcomes of the work.

The This project proposed, implemented, and evaluated a comprehensive explainable machine learning framework for the early detection of neurodevelopmental disorders in children using the 2024 National Survey of Children's Health dataset. The framework was designed to address the growing burden of NDDs in the pediatric population by tackling the key challenges of high dimensionality, class imbalance, model opacity, and the absence of actionable clinical outputs within a single unified pipeline.

Data Foundation and Preprocessing

The 2024 NSCH dataset, comprising 51,375 parent-reported child health records with 457 survey features, was selected as the primary data source due to its national representativeness and breadth of health indicators. A binary NDD target label was constructed by combining six individual disorder columns covering ADHD, Autism Spectrum Disorder, Developmental Delay, Intellectual Disability, Learning Disability, and Speech Disorder, producing 11,775 positive cases representing 22.6 percent of the dataset. Data leakage was prevented by removing all NDD diagnosis sub-columns, missing values were handled through median imputation preserving all 51,375 records, and the dataset was divided using an 80 to 20 stratified split with a fixed random seed of 42, producing a training set of 41,100 samples and a test set of 10,275 samples.

Feature Selection

A three-stage feature selection pipeline reduced 432 numeric features to a compact and clinically meaningful set of 40 variables. Stage 1 applied Variance Threshold filtering, reducing features from 432 to 371. Stage 2 applied Chi-Square testing to retain the top 80 statistically associated features. Stage 3 applied Random Forest importance ranking to select the final 40 features. The top features identified, including functional difficulty, need for extra medical care, memory conditions, and special healthcare needs status, aligned closely with established clinical knowledge about NDD risk factors, validating the pipeline's clinical relevance.

Class Imbalance Handling

The training set exhibited a significant class imbalance with 31,680 negative samples and 9,420 positive samples. SMOTE was applied exclusively to the training set after the train-test split, generating 22,260 synthetic NDD-positive samples to produce a balanced training set of 63,360 samples. This methodologically correct application avoided data leakage into the test set, ensuring all performance metrics reflect genuine generalization to real unseen survey responses.

Model Training and Evaluation

Four base classifiers were trained on the balanced training data: Logistic Regression, Random Forest, XGBoost, and Support Vector Machine. XGBoost achieved the highest individual performance with 90.11 percent accuracy and an AUC of 0.942. Random Forest followed with 89.95 percent accuracy and an AUC of 0.939, while Logistic Regression and SVM achieved 88.84 and 88.87 percent accuracy respectively. All four models substantially exceeded the 77.1 percent majority-class baseline, confirming genuine discriminative learning across all classifiers.

Stacking Ensemble

A stacking ensemble was constructed using all four base classifiers as the first layer and Logistic Regression as the meta-learner, trained with five-fold cross-validation. The ensemble achieved 89.60 percent accuracy and an AUC of 0.937 with a more balanced precision-recall profile than any individual model. Critically, the stacking model achieved 77 percent recall for NDD-positive cases compared to XGBoost's 74 percent, reducing false negatives from 604 to 536 and correctly identifying 68 additional children with NDD, demonstrating the clinical value of ensemble combination for balancing sensitivity and specificity.

Explainability and Risk Stratification

SHAP TreeExplainer was applied to the XGBoost model, generating a global summary plot identifying the most influential features across the test population. The top SHAP predictors aligned directly with established clinical knowledge, validating the model's learned relationships. The risk stratification module categorized all test set children into low, medium, and high risk tiers. The high-risk tier of 2,153 children contained 1,739 actual NDD cases, representing an 80.8 percent detection rate, while the low-risk tier of 7,539 children contained only 140 NDD cases, an NDD rate of just 1.9 percent.

Interactive Clinical Tools and Disorder-Specific Analysis

An interactive clinical predictor was developed using ipywidgets, alongside a behavioral screening questionnaire providing ADHD, Autism, and Developmental Delay subscores. Disorder-specific models revealed that individual disorder detection is substantially more challenging than combined NDD detection, with ADHD achieving only 63.73 percent accuracy and an F1-score of 0.18, confirming that the combined approach is more appropriate for population-level screening.

This project delivered a complete, reproducible, and clinically validated machine learning framework that addresses the key challenges of NDD detection simultaneously, representing a meaningful contribution to pediatric health informatics.

6.2 Limitations

Despite Despite the strong performance demonstrated by the proposed framework, several important limitations must be acknowledged to properly contextualize the findings and guide future research.

1. Parent-Reported Data and Reporting Bias The entire dataset is based on responses provided by parents or primary caregivers rather than clinically confirmed assessments. Parents may not be fully aware of their child's developmental profile, and social desirability bias may influence how sensitive conditions such as behavioral problems or mental health diagnoses are reported. The ground truth labels therefore reflect parent-perceived diagnoses rather than independently verified clinical findings, introducing uncertainty that cannot be corrected through modeling alone.

2. Geographic Generalizability The NSCH dataset is limited to children living in the United States, and the model was trained on data reflecting US demographic, socioeconomic, and healthcare system characteristics. Direct generalization to other countries, particularly India and other low and middle income nations with fundamentally different healthcare infrastructure, cultural contexts, and NDD prevalence patterns, would require significant adaptation and validation on local data before clinical deployment.

3. Binary Combined Label The framework treats any NDD diagnosis as a single positive class, combining six distinct disorders into one binary label. While this approach achieves high overall accuracy, it cannot distinguish between specific disorders. A child flagged as high risk could have ADHD, Autism, a Learning Disability, or any combination of the six NDDs. The disorder-specific models demonstrated that individual disorder detection from general survey data alone remains a substantially harder problem, with F1-scores below 0.20 for all three disorders tested, confirming that the framework is a screening tool rather than a diagnostic instrument.

4. Limited Interactive Tool Coverage The interactive clinical predictor uses only five input features for simplicity, which may not adequately capture the full clinical picture for all patients. A more comprehensive tool would require input across all 40 selected features. Additionally, the tool is currently implemented as a Google Colab widget and is not accessible as a standalone web application that practitioners could use in a real clinical setting without technical knowledge of the Colab environment.

5. Absence of Clinical Biomarkers The NSCH dataset does not include clinical measurements such as neuroimaging, standardized psychological assessments, genetic data, or developmental milestone records. The model relies entirely on parent-reported behavioral and healthcare observations, which are inherently subjective. This represents a ceiling on achievable sensitivity and specificity that cannot be overcome through algorithmic improvements without incorporating additional data modalities.

6. Single Year of Data The model was trained and evaluated exclusively on the 2024 NSCH dataset. Cross-year and cross-dataset validation was not conducted due to data availability constraints, meaning the temporal stability of the model's learned relationships and its performance on data from different survey years remains unverified.

6.3 Future Scope

The limitations identified above, combined with the strengths of the proposed framework, define several promising and practically important directions for future research and development. The following points outline the most impactful opportunities to extend the reach, accuracy, and clinical utility of the framework.

1. Cross-Population Validation

- The most immediate priority is validating the framework on health survey data from other countries, particularly India, where NDD screening infrastructure remains critically underdeveloped and the shortage of trained developmental specialists is most severe.
- India, with a population of 1.4 billion people and a significant shortage of developmental pediatricians and child psychiatrists, represents the highest-priority target for cross-population adaptation of this framework.
- Adapting the pipeline to India's National Family Health Survey or similar instruments would test whether the 40 features identified through the NSCH analysis generalize across fundamentally different cultural, socioeconomic, and healthcare contexts.
- It is possible that different features emerge as the most predictive in other national contexts, reflecting differences in how developmental conditions manifest in parent reporting, how healthcare is accessed, and how NDD diagnoses are made at the community level.
- Cross-population validation would identify which components of the pipeline, including the feature selection approach, SMOTE methodology, and stacking architecture, are truly universal and which require local adaptation before deployment.
- Collaboration with public health organizations, national health ministries, and academic institutions in target countries would be essential for accessing locally representative data and ensuring the adapted framework aligns with local clinical workflows.

2. Improving Disorder-Specific Detection

- The poor F1-scores achieved by the individual disorder models, particularly ADHD at 0.18 and Autism at 0.06 for the positive class, represent the most technically challenging open problem identified by this research and a critical priority for future work.
- Cost-sensitive learning frameworks that assign asymmetric misclassification costs to false negatives and false positives for each specific disorder would directly address the clinical reality that missing a rare disorder is substantially more harmful than generating a false alarm.

- Few-shot learning approaches, specifically designed to learn effective classifiers from very limited positive examples, represent a promising direction for detecting rare NDDs like Intellectual Disability, which had only 697 positive cases in the full dataset.

- Multi-label classification architectures that predict the presence or absence of each NDD simultaneously, rather than treating them as independent binary problems, could leverage the well-documented co-occurrence patterns between disorders to improve individual detection accuracy.

- Disorder-specific feature engineering, drawing on detailed clinical domain knowledge to construct targeted features capturing the specific behavioral and developmental profiles of each individual disorder, could substantially improve performance beyond what general-purpose data-driven feature selection achieves.

3. Multimodal Data Integration

- Combining parent-reported survey data with standardized clinical assessments, electronic health record data, school performance records, or wearable sensor data could substantially improve both sensitivity and specificity beyond what survey data alone can achieve.

- Standardized clinical tools such as the Ages and Stages Questionnaire or the Modified Checklist for Autism in Toddlers, which are already routinely administered in many pediatric settings, could be linked to survey data to provide validated behavioral screening scores alongside parent-reported indicators.

- School performance records, including standardized test scores, attendance data, and teacher behavioral reports, capture the educational functional impairment that is central to many NDD diagnoses but only partially reflected in parent survey responses.

- Wearable sensor data that objectively measures activity levels, sleep patterns, and social interaction indicators could add a behavioral measurement dimension without requiring specialist clinical involvement.

- Attention-based deep learning fusion architectures could combine structured survey features with unstructured clinical notes processed through natural language processing, capturing complementary information from multiple data sources simultaneously.

- Federated learning approaches that train on multimodal data distributed across hospitals, schools, and community health centers without centralizing sensitive patient data would address privacy concerns while enabling richer and more accurate models.

4. Deployable Clinical Web Application

- The interactive clinical predictor developed in this study as a Google Colab widget represents a proof of concept that must be extended into a fully deployable web application accessible from any browser-enabled device before real-world clinical impact can be achieved.

- A production-ready clinical tool would need a comprehensive and clinician-friendly input form covering all 40 selected features, with survey variable codes translated into plain-language clinical descriptions that practitioners can understand without technical knowledge of the NSCH instrument.

- Automatic SHAP explanation generation for each individual prediction, presented visually alongside the risk category and probability score, would allow clinicians to evaluate the model's reasoning and communicate it effectively to families.

- Integration with electronic health record systems through standard healthcare interoperability protocols would allow automatic pre-filling of known patient data, reducing data entry burden and enabling seamless incorporation into routine well-child visit workflows.

- A clinical feedback mechanism allowing practitioners to record whether model predictions were subsequently confirmed or disconfirmed by formal clinical assessment would create a continuous learning loop, enabling the model to improve over time based on real-world deployment data.

5. Longitudinal and Temporal Analysis

- The current study trained and evaluated the framework on a single year of NSCH data, leaving important questions about temporal stability and prospective predictive validity unanswered that must be addressed before clinical deployment.

- Training the framework on multiple consecutive years of NSCH data would establish whether the identified predictors and the model's learned relationships remain stable over time as healthcare practices, diagnostic criteria, and population characteristics evolve.

- Cross-year validation, where a model trained on one year's data is evaluated on the subsequent year's survey, would provide a realistic estimate of how the framework would perform in a genuine prospective deployment scenario.
- Linking survey responses across years for individual children, where permitted by privacy protocols, would enable prospective analysis of whether early survey-based risk scores predict later clinical diagnoses, which is the ultimate test of the framework's value as an early warning system.
- Time-series analysis of NDD prevalence trends across multiple NSCH years could provide valuable public health insights into the drivers of increasing NDD diagnosis rates, informing prevention and early intervention policy at the national level.

6. Advanced Explainability and Fairness Analysis

- Counterfactual explanations, which identify the minimum changes to a child's feature values that would shift their predicted risk from high to low, would provide directly actionable clinical guidance by identifying specific modifiable risk factors that could be targeted through early intervention.
- For example, a counterfactual explanation might indicate that addressing a child's unmet healthcare access needs would reduce their NDD risk probability below the high-risk threshold, giving clinicians and families a concrete and actionable intervention target.
- Concept-based explanations that map individual model features to higher-level clinical concepts, rather than raw survey variable codes, would make the model's reasoning more accessible to practitioners without deep familiarity with the NSCH instrument.
- Uncertainty quantification methods such as conformal prediction or Bayesian approaches could provide calibrated confidence intervals around each risk prediction, allowing practitioners to distinguish reliably between high-confidence classifications and borderline cases where additional assessment is most warranted.
- Fairness analysis using SHAP should be systematically conducted across demographic groups defined by race, ethnicity, household income, insurance status, and geographic region to ensure that the framework's predictions are equitable and do not systematically disadvantage already underserved populations.

CHAPTER-7

7. REFRENCES

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